train input, full runs — small-4way

L1D 16 sets x 4, L2 64 sets x 8 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

31 of 32 runs completed, 1 running, 0 pending; **1 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	21.33%	51.32%	55.49%	21.43%	51.50%	55.81%
parest	1/1	0.18%	0.45%	1.28%	0.49%	0.93%	7.38%
deepsjeng	1/1	2.99%	9.41%	15.98%	8.19%	23.80%	48.29%
leela	1/1	0.43%	1.55%	2.04%	6.08%	15.99%	46.72%
namd	1/1	3.22%	8.11%	17.74%	3.23%	8.13%	17.81%
lbm	1/1	9.69%	27.97%	35.99%	9.69%	27.97%	35.99%
nab	1/1	7.61%	25.91%	37.53%	8.29%	27.38%	44.95%
nab_gcn4dna	1/1	7.30%	24.89%	35.20%	8.25%	26.98%	45.65%
xz	1/1	1.36%	4.25%	4.82%	6.14%	16.52%	26.27%
xz_combined	1/1	2.79%	10.48%	25.95%	4.09%	14.43%	41.80%
imagick	1/1	0.04%	0.21%	4.19%	0.25%	0.82%	48.78%
blender	1/1	0.31%	0.55%	4.03%	3.18%	5.93%	39.04%
povray	1/1	0.27%	0.46%	0.95%	14.26%	22.21%	55.40%
omnetpp	1/1	0.01%	0.02%	0.03%	12.59%	24.64%	48.38%
xalancbmk	1/1	0.44%	1.00%	1.04%	18.29%	32.42%	63.05%
x264	1/1	6.30%	13.42%	40.78%	8.75%	18.41%	57.31%
gcc_200	1/1	1.16%	2.64%	4.34%	16.68%	35.31%	67.49%
gcc_scilab	1/1	1.25%	2.90%	4.70%	15.74%	33.65%	65.08%
gcc	1/1	1.41%	3.09%	5.21%	16.29%	32.84%	65.64%
mcf	1/1	1.61%	2.27%	13.28%	2.75%	4.04%	21.88%
perlbench_scrabbl	1/1	0.23%	0.30%	0.93%	17.65%	32.90%	56.14%
perlbench_suns	1/1	0.98%	2.06%	2.86%	18.85%	36.45%	59.85%
perlbench_diffmail	1/1	2.06%	4.51%	4.76%	17.54%	31.15%	53.97%
perlbench_perfect	1/1	1.44%	2.34%	5.00%	18.85%	33.31%	60.43%
perlbench_splitmail	1/1	5.79%	9.36%	20.68%	17.56%	30.55%	58.86%
bwaves_1	1/1	2.99%	7.65%	18.86%	2.99%	7.65%	18.86%
bwaves_2	1/1	3.04%	7.82%	18.83%	3.04%	7.82%	18.83%
exchange2	1/1	4.84%	10.67%	12.41%	6.03%	13.09%	15.72%
fotonik3d	1/1	3.99%	10.66%	12.65%	4.00%	10.69%	12.75%
roms	0/1	—	—	—	—	—	—
wrf	1/1	0.92%	3.44%	2.61%	1.59%	4.97%	8.09%
cam4	1/1	2.49%	9.15%	7.96%	4.59%	14.65%	19.40%
pooled		3.60%	9.44%	17.30%	7.34%	17.15%	41.91%
mean		3.18%	8.35%	13.49%	9.59%	20.87%	41.47%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.30%	4.85%	34.48%	4.27%	12.94%	65.68%	12.63%	8.09%
parest	done	10.27%	2.03%	10.97%	1.46%	1.95%	69.50%	3.10%	14.83%
deepsjeng	done	8.71%	2.02%	11.23%	7.46%	4.53%	7.25%	2.38%	13.25%

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
leela	done	5.26%	1.74%	5.99%	0.20%	3.26%	20.07%	0.00%	85.89%
namd	done	11.59%	2.64%	14.71%	1.23%	2.49%	28.04%	8.63%	18.29%
lbm	done	21.30%	0.00%	7.69%	0.50%	73.99%	97.61%	—	29.39%
nab	done	5.48%	0.42%	8.59%	0.27%	0.59%	5.51%	0.15%	1.23%
nab_gcn4dna	done	4.91%	0.24%	7.65%	0.35%	0.51%	58.63%	6.67%	2.57%
xz	done	6.00%	0.87%	8.78%	2.99%	1.65%	38.14%	0.00%	3.07%
xz_combined	done	9.82%	1.75%	12.52%	0.81%	4.50%	23.02%	0.04%	2.16%
imagick	done	5.01%	34.95%	5.00%	0.48%	1.76%	94.98%	1.01%	4.90%
blender	done	6.71%	5.33%	7.30%	4.78%	2.85%	12.80%	1.66%	9.49%
povray	done	9.02%	6.99%	10.95%	19.46%	4.08%	46.64%	67.66%	72.00%
omnetpp	done	15.94%	0.05%	21.32%	0.83%	5.39%	29.81%	97.14%	3.02%
xalancbmk	done	11.88%	15.33%	15.40%	26.56%	4.71%	12.81%	0.00%	0.04%
x264	done	1.97%	0.22%	2.41%	0.88%	1.87%	32.24%	2.56%	15.02%
gcc_200	done	8.32%	1.20%	11.14%	4.23%	3.34%	41.08%	4.22%	9.44%
gcc_scilab	done	8.90%	1.48%	11.93%	3.91%	3.52%	41.20%	2.79%	10.33%
gcc	done	8.69%	1.46%	11.60%	4.12%	3.38%	43.73%	3.52%	9.80%
mcf	done	23.17%	5.26%	26.93%	7.39%	7.33%	77.15%	6.07%	18.62%
perlbench_scrabbl	done	6.20%	3.29%	8.86%	8.91%	1.65%	14.10%	0.00%	0.01%
perlbench_suns	done	5.44%	0.39%	7.71%	8.94%	1.73%	20.70%	0.26%	12.21%
perlbench_diffmail	done	4.67%	0.16%	6.45%	2.76%	1.85%	41.21%	0.21%	22.43%
perlbench_perfect	done	8.25%	0.67%	11.52%	5.52%	2.71%	55.17%	39.20%	25.84%
perlbench_splitmail	done	4.50%	0.17%	6.51%	2.96%	1.69%	19.98%	0.42%	2.44%
bwaves_1	done	43.00%	4.98%	47.49%	0.14%	40.88%	26.58%	3.92%	0.32%
bwaves_2	done	42.86%	4.85%	47.39%	0.14%	41.29%	27.02%	3.99%	0.32%
exchange2	done	1.85%	1.40%	2.44%	2.35%	1.02%	1.76%	0.26%	0.87%
fotonik3d	done	11.67%	3.81%	15.96%	0.01%	0.20%	59.27%	1.90%	20.68%
roms	running	—	—	—	—	—	—	—	—
wrf	done	11.05%	6.76%	11.47%	4.70%	10.29%	34.94%	8.73%	12.40%
cam4	done	10.12%	1.77%	10.55%	2.13%	11.66%	59.86%	11.58%	31.83%
pooled		10.27%	2.96%	12.53%	3.12%	6.51%	41.62%	9.53%	9.65%
mean		11.22%	3.78%	13.64%	4.22%	8.38%	38.92%	9.69%	14.86%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?

column	formula	aciklama
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.